Selected Perception Methods

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30 October 2026

The `perception_challenge_best` branch retains the selected stereo hold estimator, configured robot estimator and persistent RGB robot/cup tracker. It contains one trained checkpoint, one example per challenge, isolated setup and camera calibration. Alternatives and historical experiments remain on `perception_challenge_full`. The command reference is `docs/perception/README.md`.

1 Retained methods

Stereo holds. Color proposals initialize dual-view exact CAD fitting. A verified public-generator search can recover an untouched native layout; failed verification retains the independent CAD estimate. Input is a stereo PNG and declared rig calibration. Output is `holds.json`: CAD backplate anchors in metres, intrinsic XYZ Euler radians, hold type and color. Exact recovery assumes ten Microban holds, zero to three volumes and the released generator geometry/appearance. It does not apply to camera photographs.

Saved robot frames. A 20-point Microban network initializes the known joint configuration. M3T tracks the configured links; protected CAD refinement and a trimmed boundary fallback estimate the trunk transform. The selected `max` profile requires two RGB images, calibrated intrinsics, synchronized timestamps and all 19 measured joints. It does not reconstruct unknown joints.

Continuous perception. The `fast` profile retains persistent M3T, a dense RGB cup map and conditional socket correction. Candidate sockets must have image support and satisfy geometry gates. Socket proximity does not confirm seating or loading. Cup geometry and lattice priors require physical measurement.

2 Install and reproduce

Use Linux x86-64, Python 3.12, Git/Git-LFS, C++ tools, EGL/OpenGL and CUDA 12.8. The tested GPU is RTX 5000 Ada, compute capability 8.9; change `-cuda-arch` during setup for another supported card.

```
git submodule update --init
python3.12 scripts/perception/setup.py --robot --cuda
source .venv-perception/bin/activate
python scripts/perception/run.py doctor --robot
python scripts/perception/run.py reproduce --out outputs/holds
python scripts/perception/run.py replay \
  --dataset data/perception/robot --out outputs/robot
python scripts/perception/verify_replays.py \
  --dataset data/perception/robot --predictions outputs/robot \
  --out outputs/robot/verification.json
```

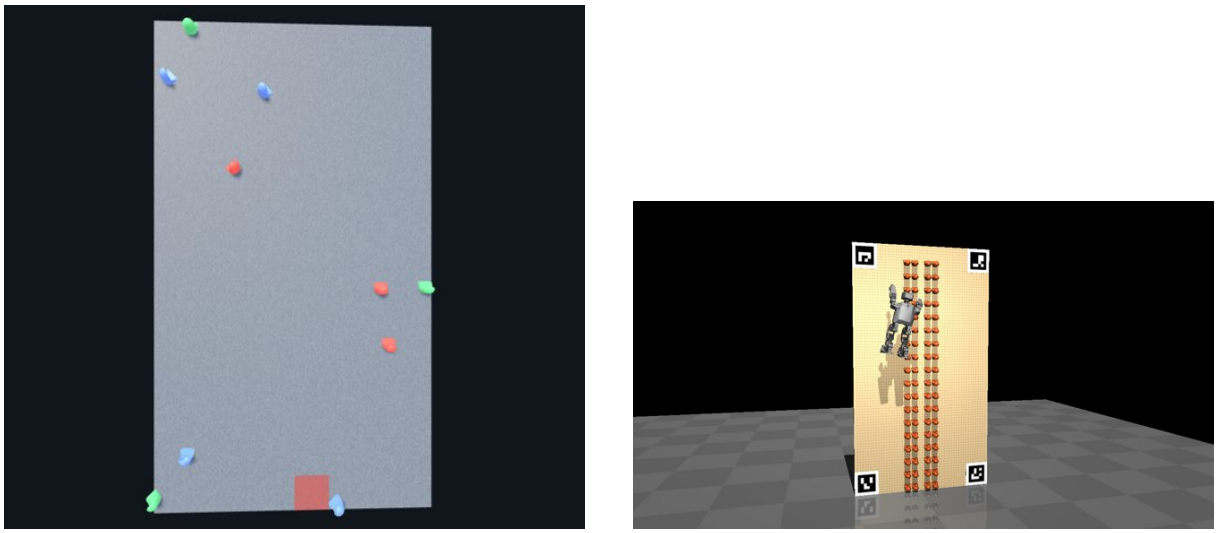


Figure 1: The included stereo hold scene (left view) and configured robot scene (first camera). Inference uses both views. Saved examples are regression cases.

Setup creates a private environment and cache. Torch/CUDA/native dependencies still require several GB. GPU jobs wait for spare capacity and poll every 30 minutes while externally busy. The runtime does not stop other jobs.

3 Cameras and measurements

`capture -list` enumerates devices. Select two serials, capture RGB pairs, then run `calibrate` on varied printed-board poses. The guide gives the complete commands. Only color streams are enabled. Intrinsic calibration fits SDK-rectified frames and is bound to serial order, resolution and source intrinsics. Wall pose is estimated separately from markers or board boundaries.

Supply a measured encoder JSONL log with `clock: host_monotonic`, `timestamp_s` and all 19 `joints_rad`. Map acquisition times into the camera laptop's clock; an independent Pi clock or receipt time is insufficient.

```
python scripts/perception/run.py live \
  --serial LEFT_SERIAL --serial RIGHT_SERIAL \
  --telemetry-log outputs/encoders.jsonl \
  --camera-calibration outputs/calibration/calibration.json \
  --out outputs/live
```

Live defaults to combined robot/cup tracking. Use `-task robot` for root pose alone. Consumers read `latest.json` and require `valid` and `fresh`. Poses older than the default 0.25 s limit are suppressed. Inspect map timestamps and refresh errors separately; reacquisition may take seconds.

The private model reproduces the original simulator. Team main uses the cut-foot fork. Physical fitting must use the measured geometry through `RCR_MICROBAN_MODEL`. The guide explains this override and team frame conversion. A complete team wall model still needs verified start contacts.

4 Checks and scope

After removing alternatives, the selected code reproduced all 50 source hold records on the five saved stereo cases from the full branch. All five robot cases passed the local root thresholds

(5 mm and 5 degrees) and the 5 mm bound for every endpoint. Combined replay produced a valid robot pose and 72 mapped cup candidates on the retained example, with no refresh error. Candidate count is not an independently scored map-accuracy result.

The repository carries only one case per challenge; the compact `verification.json` records the broader saved-case check. References are opened after inference. These are regression checks, not new blind benchmarks. Calibration and timing paths have software tests. No physical cameras were attached, so USB reliability, exposure synchronization, physical accuracy and a camera-driven climb remain unvalidated. The legacy images-only team functions remain stubs; use the explicit packet API to supply calibration and telemetry.